

Tony Chen

tonychengangyi@gmail.com | github.com/tonychen-2006 | tony-chen.dev

EDUCATION

The University of British Columbia

Bachelor of Applied Science - Computer Engineering

Vancouver, BC

Expected May 2028

Relevant Coursework: Software Construction, Computing Systems, Algorithms and Data Structures, Circuit Analysis

TECHNICAL SKILLS

Programming Applications: C, C++, Java, Python, SystemVerilog, Assembly, MATLAB, Simulink

Embedded Systems: FreeRTOS, Zephyr RTOS, Embedded Linux, CMake, Device Drivers, Interrupt Systems

Communication Protocols: CAN, UART, I2C, DMA, LoRa, BLE

Hardware and Debugging Systems: STM32, ESP32, Logic Analyzer, Oscilloscope, J-Link, GDB, Makefile

WORK EXPERIENCE

Verdi

Firmware Developer Intern

Vancouver, BC

Jan 2026 – Apr 2026

- Led migration of embedded C firmware from FreeRTOS to Zephyr RTOS, re-architecting task scheduling and driver interfaces, adapting CMake build systems and device tree overlays for scalable multi-board support.
- Implemented micro-valve control firmware using GPIO and RTOS threads, enabling automated valve actuation with safe boot, continuous state recovery, and watchdog-supervised real-time operation.
- Integrated and validated LoRa communication (RAK1337), implementing initialization, packet handling, and RTOS-safe asynchronous message processing.
- Designed and implemented a CI/CD pipeline for pull requests using GitHub Actions, triggering builds and unit tests based on board-specific file changes, and enforcing code quality with automated clang-format checks.

EIM Technologies

Embedded Systems Engineer Intern

Richmond, BC

May 2026 – Aug 2026

- Developing STM32 firmware for a real-time audio DSP platform, implementing AUX/ADC signal acquisition, DMA circular buffering, FFT spectrum analysis, and FIR/IIR digital filtering.
- Designed hardware and firmware workflows for analog input conditioning, sampling-rate control, SPI display visualization, and audio signal reconstruction.
- Prototyping a custom STM32-based logic analyzer with high-speed GPIO sampling, trigger-based capture, timestamped signal recording, and UART/I2C/SPI debugging support.

UBC Solar Racing (Student Design Team)

Senior Embedded Systems Designer

Vancouver, BC

Sep 2024 – Present

- Led firmware development for the Driver Dashboard subsystem, implementing vehicle drive-state control logic as a safety-critical state machine while integrating CAN bus messaging and hardware abstraction layers.
- Built a Raspberry Pi-based Python gateway connected to a Netgear hotspot to parse vehicle telemetry data and stream it to InfluxDB/Grafana, reducing latency (< 2 seconds) and enabling real-time remote monitoring.
- Implemented a common STM32 CAN bootloader prototype for vehicle boards, supporting application jump logic, firmware image transfer, CRC validation, flash programming, and Raspberry Pi-based .bin deployment tooling.
- Developed Telemetry Board PCB firmware supporting cellular and radio communication paths, packaging CAN messages with timestamps, CAN IDs, payload length, and UART framing while implementing whitelist-based filtering and rate-limiting to reduce unnecessary bandwidth.
- Implemented GPS data acquisition firmware as a FreeRTOS thread that parsed NMEA sentences (GGA, GLL, GSA, RMC, VTG, GSV), validated checksums, and packaged results into CAN frames.
- Designed and implemented a FreeRTOS-based cruise control system, using a custom PI controller with real-time force modeling (drag, rolling resistance, hill climbing) for precise speed regulation, fusing IMU and CAN data with a Kalman filter, and ensuring critical safety logic conditions.

PROJECTS

Linux Home Server and Router Infrastructure

Jun 2026 – Present

- Building a Linux home server with a Lenovo ThinkCentre for self-hosted storage, monitoring dashboards, home automation control, secure VPN access, and local service hosting.
- Configuring SSH, Tailscale, DNS/DHCP, firewall rules, router-side management, and dashboards for network visibility and smart-home device control.